

Slawinski

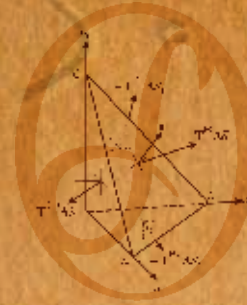
WAVES AND RAYS IN ELASTIC CONTINUA

"Professor Slawinski has written an introductory textbook that rigorously develops the foundations for elastodynamics and wave propagation in anisotropic elastic media in a clear and well written style. The main text is supplemented with many worked examples and suggestions for further reading. This text would be a useful textbook for a senior undergraduate or introductory graduate level, applied mathematics course in waves in elastic media, and as an introduction to research in theoretical seismology."

Chris Chapman, Cambridge University

"This impressive treatise contains a lucid and comprehensive description of material symmetry in the context of seismic wave propagation. These insights form the foundation of a careful and detailed analysis of seismic waves in anisotropic media. 'Waves & Rays' is very well-suited as a text for a graduate course and as a reference monograph for experts in the field."

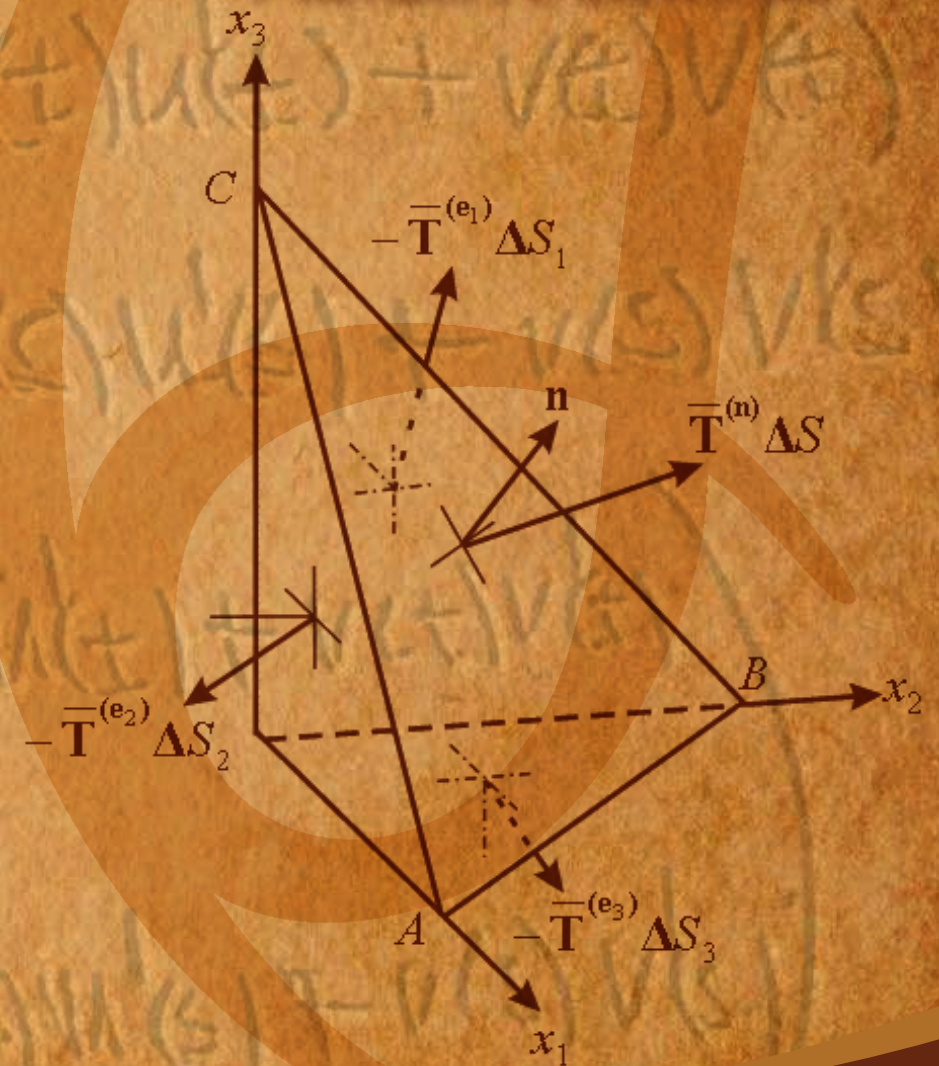
Jeroen Tromp, Princeton University



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Michael A. Slawinski



Preface

Il ne suffit pas d'observer, il faut se servir de ses observations, et pour cela il faut généraliser. [...] Le savant doit ordonner; on fait la science avec des faits comme une maison avec des pierres; mais une accumulation de faits n'est pas plus une science qu'un tas de pierres n'est une maison.¹

Henri Poincaré (1902) La science et l'hypothèse

Theoretical formulations of applied seismology are substantiated by observable phenomena. Reciprocally, our perception and understanding of these phenomena necessitate rigorous descriptions of physical behaviours. As stated by Bunge in his book on "Philosophy of Science, Vol. I: From problem to theory",

A nice illustration of the intertwining of empirical and theoretical events in the actual practice of science is offered by seismology, the study of elastic disturbances of Terra. [...] In conclusion, in order to "read" a seismogram so that it may become a set of data regarding an event (e.g., an

¹ It is not enough to observe. One must use these observations, and for this purpose one must generalize. [...] The scientist must organize [knowledge]; science is composed of facts as a house is composed of bricks; but an accumulation of facts is no more a science than a pile of bricks is a house.

To emphasize this statement of Poincaré, let us also consider the following quotation.

As the biggest library if it is in disorder is not as useful as a small but well-arranged one, so you may accumulate a vast amount of knowledge but it will be of far less value to you than a much smaller amount if you have not thought it over for yourself; because only through ordering what you know by comparing every truth with every other truth can you take complete possession of your knowledge and get it into your power.

Arthur Schopenhauer (1851) Parerga and Paralipomena, Volume 2

earthquake) or an evidence relevant to a theory (e.g., about the inner structure of our planet), the seismologist employs elasticity theory and all the theories that may enter the design and interpretation of the seismograph.

The present book emphasizes the interdependence of mathematical formulation and physical meaning in the description of seismic phenomena. Herein, we use aspects of continuum mechanics, wave theory and ray theory to explain phenomena resulting from the propagation of seismic waves.

The book is divided into three main parts: *Elastic continua*, *Waves and rays* and *Variational formulation of rays*. There is also a fourth part, which consists of *Appendices*. In *Part 1*, we use continuum mechanics to describe the material through which seismic waves propagate, and to formulate a system of equations to study the behaviour of such a material. In *Part 2*, we use these equations to identify the types of body waves propagating in elastic continua as well as to express their velocities and displacements in terms of the properties of these continua. To solve the equations of motion in anisotropic inhomogeneous continua, we use the high-frequency approximation and, hence, establish the concept of a ray. In *Part 3*, we show that, in elastic continua, a ray is tantamount to a trajectory along which a seismic signal propagates in accordance with the variational principle of stationary traveltime. Consequently, many seismic problems in elastic continua can be conveniently formulated and solved using the calculus of variations. In *Part 4*, we describe two mathematical concepts that are used in the book; namely, homogeneity of a function and Legendre's transformation. This part also contains a *List of symbols*.

The book contains an *Index* that focuses on technical terms. The purpose of this index is to contribute to the coherence of the book and to facilitate its use as a study manual and a reference text. Numerous terms are grouped to indicate the relations among their meanings and nomenclatures. Some references to selected pages are marked in bold font. These pages contain a defining statement of a given term.

This book is intended for senior undergraduate and graduate students as well as scientists interested in quantitative seismology. We assume that the reader is familiar with linear algebra, differential and integral calculus, vector calculus, tensor analysis, as well as ordinary and partial differential

equations. The chapters of this book are intended to be studied in sequence. In that manner, the entire book can be used as a manual for a single course. If the variational formulation of ray theory is not to be included in such a course, the entire *Part 3* can be omitted.

Each part begins with an *Introduction*, which situates the topics discussed therein in the overall context of the book as well as in a broader scientific context. Each chapter begins with *Preliminary remarks*, which state the motivation for the specific concepts discussed therein, outline the structure of the chapter and provide links to other chapters in the book. Each chapter ends with *Closing remarks*, which specify the limitations of the concepts discussed and direct the reader to related chapters. Each chapter is followed by *Exercises* and their solutions. While some exercises extend the topics covered, others are referred to in the main text. Reciprocally, the footnotes attached to these latter exercises refer the reader to the sections in the main text, where a given exercise is mentioned. Often, the exercises referred to in the main text supply steps that are omitted from the exposition in the text. Also, throughout the book, footnotes refer the reader to specific sources included in the *Bibliography*.

“Seismic waves and rays in elastic media” strives to respect the scientific spirit of Rudzki, described in the following statement² of Marian Smoluchowski, Rudzki’s colleague and friend.

Tematyka geofizyczna musiała nęcić Rudzkiego, tak wielkiego, fantastycznego miłośnika przyrody, z drugiej zaś strony ta właśnie nauka odpowiadała najwybitniejszej właściwości umysłu Rudzkiego, jego dążeniu do matematycznej ścisłości w rozumowaniu.³

² Smoluchowski, M., (1916) Maurycy Rudzki jako geofizyk / Maurycy Rudzki as a geophysicist: *Kosmos*, **41**, 105 – 119

³ The subject of geophysics must have attracted Rudzki, a great lover of nature. Also, this very science accommodated the most outstanding quality of Rudzki’s mind, his striving for mathematical rigour in reasoning.

Changes from First Edition

Mathematics gives to science the power of abstraction and generalization, and a symbolism that says what it has to say with the greatest possible clarity and economy.

John Lighton Synge and Byron A. Griffith (1949)
Principles of mechanics

The conviction expressed by the above quote has inspired both the first and second editions of this book. Significant additions and modifications have been included in the second edition. These changes came about as a result of teaching from this text, readers's enquiries, and research into concepts discussed in the first edition. Summary of these changes and the reason for their inclusion are given below.

Part 1:

Chapter 1: To familiarize the reader with the fundamentals that underlie the seismic theory, we introduce in Section 1.2 the three rudimentary concepts of continuum mechanics, namely, material body, manifold of experience and system of forces.

Chapter 2: To deepen the reader's understanding of the arguments used to formulate the balance principles of continuum mechanics, we include in Sections 2.4 and 2.7 the particle-mechanics motivation of the balance of linear momentum and the balance of angular momentum, respectively. To make the reader aware of possible extensions within continuum mechanics, we discuss in Section 2.7 the distinction between the strong and weak forms of Newton's third law of motion. As a results, we show that the symmetry of the stress tensor is a consequence of the strong form, and not a fundamental physical law.

Chapter 3: To deepen the reader's understanding of the fundamentals that underlie the formulation of seismic theory, we introduce in Section 3.1 the three principles of constitutive equations, namely, determinism, local

action and objectivity. To provide the background that allows us to emphasize the meaning of elasticity, we introduce in Section 3.4 constitutive equations of anelastic continua.

Chapter 5: Following a demonstration that only eight classes of the elasticity tensor exist,⁴ all these classes are included in this edition. The addition consists of Sections 5.8 and 5.11, where we study trigonal and cubic continua.

Part 2

Chapter 6: Since this chapter deals with the wave equation, which plays an important role in seismology, we wish to deepen the reader's understanding of different aspects of this equation. In Section 6.5, we discuss well-posedness of the wave equation with its initial conditions. To provide further insight into the properties of the wave equation, we compare it in Section 6.7 to the evolution equation, which is ill-posed. Another significant addition consists of the formulation and study of solutions of the wave equation in two and three spatial dimensions, as presented in Section 6.6. Therein, we discuss also the range of influence and the domain of dependence of these solutions. To familiarize the reader with the concepts of reflection and transmission in the context of the wave equation, we study the solutions of one-dimensional scattering, which is discussed in Section 6.8. To familiarize the reader with the several-century-long debate concerning the applicability of the wave equation, which is a differential equation, to study nondifferentiable solutions, we present the theory of distributions and its application to the wave equation. The crux of this addition is Section 6.9. Furthermore, in Section 6.10, we elaborate on the concept of the reduced wave equation and Fourier's transform of the wave equation, which is used in the following chapter to discuss trial solutions of the equations of motion.

Chapter 7: Since this chapter deals with the equations of motion whose trial solutions necessitate asymptotic methods, we wish to familiarize the reader with the concept of the asymptotic series in the context of ray theory,

⁴ Readers interested in the existence of the eight symmetry classes might refer to Bóna, A., Bucataru, I., and Slawinski, M.A., (2005) Characterization of elasticity-tensor symmetries using $SU(2)$. *Journal of Elasticity* **75**(3), 267 – 289, and to Bóna, A., Bucataru, I., and Slawinski, M.A., (2004) Material symmetries of elasticity tensor. *The Quarterly Journal of Mechanics and Applied Mathematics* **57**(4), 583 – 598.

which is commonly referred to as the asymptotic ray theory. This addition is contained in Sections 7.2.3 and 7.2.4.

Chapter 8: Since Hamilton's ray equations are the basis of ray theory, we wish to deepen the reader's understanding of their meaning, as well as the meaning of their solutions. To do so, we present an analytic solution of Hamilton's ray equations, which allows us to gain both mathematical and physical insights into Hamilton's equations. Furthermore, an equivalent solution is presented using Lagrange's ray equations. Hence, this analytic solution allows us to relate the Hamiltonian and Lagrangian formulations of ray theory—two distinct approaches. The crux of this addition consists of Section 8.5.

Chapter 9: The convexity and detachment of the innermost phase-slowness surface are commonly invoked in seismology. In this edition, we show particular cases for which this surface is not detached, and we comment on the fact that detachment is not necessary to prove the convexity of the innermost surface.

Chapter 10: In this edition, we use Newton's third law of motion—rather than the second one used in the first edition—to derive in Section 10.2.1, the dynamic boundary conditions on the interface between two media. Since we are dealing with two media acting on one another, the third law lends itself more naturally than the second one for which we had to treat the interface as a layer with a vanishing thickness.

Part 3

Chapter 14: In this edition, the traveltime expression for a signal in a continuum exhibiting an elliptical velocity dependence with direction and a linear dependence with depth is valid for the entire ray: both the downgoing and upgoing segments. In the first edition, the expression was valid for a downgoing ray only.

In this edition, the chapter on Lagrange's ray equations is found at the end of *Part 2*, rather than immediately after the chapter on Hamilton's ray equations. This placement of the chapter emphasizes the physical motivation; namely, we can study the entire ray theory in the context of Hamilton's ray equations. Lagrange's ray equations, on the other hand, open a new path of study to which we devote *Part 3*.

As a result of teaching from the first edition, the second edition acquired more exercises and figures. There is also a *List of Figures* included.

With the above additions, the title of the book has been modified to emphasize the generality of the approach and the importance of the concept of continuum. Thus, “Seismic waves and rays in elastic media” became “Waves and rays in elastic continua”.

Changes from First Printing of Second Edition

Every partially true theory is as alive as the physicists who care to work it out, apply and analyze it: even classical theories continue to evolve and deserve being overhauled both for their own sake and as models of theory construction.

Mario Bunge (1967)
Foundations of physics

The significant addition in the second printing of the second edition is Section 6.4, in which we formulate the wave equations in terms of the displacement vector, not in terms of its derivatives. Therein, we address also the necessary and sufficient conditions for existence of P and S waves in isotropic Hookean solids. Another addition is Section 5.9.2, where we discuss different coordinate expressions of the same tensor, which we illustrate in Figure 5.9.1 and Exercises 5.5 and 5.6.

There are six significant modifications. In Section 3.4.3, we examine in detail the issues of viscoelasticity, with additions of Figures 3.4.1 and 3.4.2. In Section 5.3, we discuss relations between rotations in three dimensions and their expressions in terms of 6×6 matrices. In Section 6.1.4, an application of the Jacobian matrix in the context of the wave equations for P and S waves allows us to explain the role of the divergence and the curl in formulating these equations. To set the stage for this application, the last paragraph of Section 1.5 is modified. Another modification is at the end of Section 6.5.1 in which d'Alembert's approach is extended to the higher spatial dimension for waves exhibiting spherical symmetries; this modification is followed by the addition of Exercises 6.3 and 6.4. In Section 6.10.2, certain properties of Fourier's transform are clarified. The last modification is in Exercise 9.11, where the physical meaning of equality among the wavefront orientation and the ray and polarization directions is emphasized.

The second printing benefits from inclusion of references to Bóna, A., and Slawinski, M.A., (2011) *Wavefronts and rays as characteristics and asymptotics*, published by World Scientific—a book that can be viewed as a follow up of subjects discussed in Chapters 6–8, and which was published after the first printing. The figures have been refined and captions accommodated accordingly. Also, references to recent papers and books have been included and several index entries added.

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